

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 09 · Laws of Logarithms

With Answers

Questions with full worked solutions

- 47 pages
- Past papers up to January 2026

PURE 2 | TOPIC 09

Laws of Logarithms

Your classified route

20

QUESTIONS

148

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 09 | FAMILY 01

Change Of Base

WHAT THIS FAMILY ASKS

Recognise the change of base demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

7 total marks

T09

09_BASE

09_BASE_CORE

CHAPTER
Laws of Logarithms

FAMILY
Change of base

VARIANT
core family move

7 marks

WMA12/01 OCTOBER 2022
Question 10

Past-paper question

WMA12-01_2022_Oct_Q10

10. Given $a = \log_2 3$

(i) write, in simplest form, in terms of a ,

(a) $\log_2 9$

(b) $\log_2 \left(\frac{\sqrt{3}}{16} \right)$

(3)

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of a .

(4)

T09 Laws of Logarithms

09_BASE Change of base

09_BASE_CORE core family
move

WORKED SOLUTION

Question 10

WMA12/01 October 2022

WMA12-01_2022_Oct_Q10

1

Use the power rule for log 2 of 9

$$\log_2 9 = \log_2(3^2) = 2 \log_2 3 = 2a.$$

2

Split the second logarithm

$$\log_2 \left(\frac{\sqrt{3}}{16} \right) = \log_2(3^{1/2}) - \log_2(16).$$

3

Simplify in terms of a

$$\log_2(3^{1/2}) = \frac{1}{2} \log_2 3 = \frac{a}{2} \text{ and } \log_2(16) = 4, \text{ so the expression is } \frac{a}{2} - 4.$$

4

Take log base 2 of both sides

$$\text{For } 3^x \times 2^{x+4} = 6, \text{ take } \log_2 \text{ of both sides: } \log_2(3^x) + \log_2(2^{x+4}) = \log_2 6.$$

5

Rewrite each term

$$\log_2(3^x) = xa, \log_2(2^{x+4}) = x + 4, \text{ and } \log_2 6 = \log_2(2 \cdot 3) = 1 + a.$$

6

Solve for x

$$\text{So } xa + x + 4 = a + 1. \text{ Hence } x(a + 1) = a - 3, \text{ giving } x = \frac{a-3}{a+1}.$$

FINAL ANSWER

WMA12-01_2022_Oct_Q10

$$(i)(a) \ 2a. \quad (i)(b) \ \frac{a}{2} - 4. \quad (ii) \ x = \frac{a-3}{a+1}.$$

TOPIC 09 | FAMILY 02

Log Definition

WHAT THIS FAMILY ASKS

Recognise the log definition demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

7 total marks

T09

09_EXP

09_EXP_CORE

CHAPTER

Laws of Logarithms

FAMILY

Log definition

VARIANT

core family move

7 marks

WMA12/01 JANUARY 2020

Question 9

Past-paper question

WMA12-01_2020_Jan_Q09

9. (a) Sketch the curve with equation

$$y = 3 \times 4^x$$

showing the coordinates of any points of intersection with the coordinate axes.

(2)

The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P .

- (b) Show that the x coordinate of P is $\frac{\log_{10} 2}{\log_{10} 24}$

(5)

T09 Laws of Logarithms

09_EXP Log definition

09_EXP_CORE core family
move

WORKED SOLUTION

Question 9

WMA12/01 January 2020

WMA12-01_2020_Jan_Q09

1

Describe the sketch

$y = 3 \times 4^x$ is an increasing exponential curve. At $x = 0$, $y = 3$, so the curve crosses the y -axis at $(0, 3)$ and has no x -axis intersection.

2

Equate the two curves

At P , $6^{1-x} = 3 \times 4^x$.

3

Take logarithms

$$(1-x) \log_{10} 6 = \log_{10} 3 + x \log_{10} 4.$$

4

Collect the x terms

$$\log_{10} 6 - \log_{10} 3 = x(\log_{10} 4 + \log_{10} 6).$$

5

Simplify the logarithms

$$\log_{10} 6 - \log_{10} 3 = \log_{10} 2 \text{ and } \log_{10} 4 + \log_{10} 6 = \log_{10} 24.$$

6

State the x coordinate

$$\text{Therefore } x = \frac{\log_{10} 2}{\log_{10} 24}.$$

FINAL ANSWER

WMA12-01_2020_Jan_Q09

(a) **Increasing exponential curve through $(0, 3)$, with no x -axis intersection.**

(b) $x = \frac{\log_{10} 2}{\log_{10} 24}.$

TOPIC 09 | FAMILY 03

Log Laws

WHAT THIS FAMILY ASKS

Recognise the log laws demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

3 classified questions

20 total marks

T09

09_LAWS

09_LAWS_CORE

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7 marks

WMA12/01 OCTOBER 2019
Question 7

Past-paper question

WMA12-01_2019_Oct_Q07

7. Given $\log_a b = k$, find, in simplest form in terms of k ,

(i) $\log_a \left(\frac{\sqrt{a}}{b} \right)$ (2)

(ii) $\frac{\log_a a^2 b}{\log_a b^3}$ (2)

(iii) $\sum_{n=1}^{50} (k + \log_a b^n)$ (3)

T09 Laws of Logarithms

09_LAWS Log laws

09_LAWS_CORE core family
move

WORKED SOLUTION

Question 7

WMA12/01 October 2019

WMA12-01_2019_Oct_Q07

1

Use the given logarithm

Since $\log_a b = k$, we can replace every $\log_a b$ by k .

2

Simplify part (i)

$$\log_a \left(\frac{\sqrt{a}}{b} \right) = \log_a (a^{1/2}) - \log_a b = \frac{1}{2} - k.$$

3

Simplify the numerator and denominator

$$\log_a (a^2 b) = \log_a a^2 + \log_a b = 2 + k, \text{ and } \log_a (b^3) = 3 \log_a b = 3k.$$

4

Write part (ii)

$$\text{Therefore } \frac{\log_a (a^2 b)}{\log_a (b^3)} = \frac{k+2}{3k}.$$

5

Simplify the summand

$$\log_a (b^n) = n \log_a b = nk, \text{ so } k + \log_a (b^n) = k + nk = (n+1)k.$$

6

Sum the series

$$\sum_{n=1}^{50} (n+1)k = k \left(\sum_{n=1}^{50} n + 50 \right) = k(1275 + 50) = 1325k.$$

FINAL ANSWER

WMA12-01_2019_Oct_Q07

$$(i) \frac{1}{2} - k. \quad (ii) \frac{k+2}{3k}. \quad (iii) 1325k.$$

T09

09_LAWS

09_LAWS_CORE

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8 marks

WMA12/01 OCTOBER 2022
Question 4

Past-paper question

WMA12-01_2022_Oct_Q04

4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \quad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

- (a) show that $a = 1.072$ correct to 3 decimal places.

(3)

Using $a = 1.072$

- (b) find an equation for t in terms of w

(3)

- (c) find the value of t when $w = 15$, giving your answer to 3 significant figures.

(2)

T09 Laws of Logarithms

09_LAWS Log laws

09_LAWS_CORE core family
move

WORKED SOLUTION

Question 4

WMA12/01 October 2022

WMA12-01_2022_Oct_Q04

1

Substitute the given informationWhen $t = 3$, $w = 10$. So $10 = \log_a(3 + 5) - \log_a 4 = \log_a 8 - \log_a 4$.

2

Combine the logarithms $\log_a 8 - \log_a 4 = \log_a \left(\frac{8}{4}\right) = \log_a 2$, so $10 = \log_a 2$.

3

Find a $10 = \log_a 2$ means $a^{10} = 2$, hence $a = 2^{1/10} = 1.071773... = 1.072$ to 3 decimal places.

4

Start from the modelUsing $a = 1.072$, $w = \log_{1.072}(t + 5) - \log_{1.072} 4$.

5

Combine the logs again $w = \log_{1.072} \left(\frac{t+5}{4}\right)$.

6

Make t the subjectChanging from logarithmic form to exponential form gives $(1.072)^w = \frac{t+5}{4}$, so $t = 4(1.072)^w - 5$.

7

Substitute w equals fifteenWhen $w = 15$, $t = 4(1.072)^{15} - 5 = 6.3496...$, so $t = 6.35$ months to 3 significant figures.**FINAL ANSWER**

WMA12-01_2022_Oct_Q04

(a) $a = 1.072$ to 3 d.p. (b) $t = 4(1.072)^w - 5$. (c) $t = 6.35$ months to 3 s.f.

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09_LAWS

09_LAWS_CORE

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5 marks

WMA12/01 OCTOBER 2024

Question 6

Past-paper question

WMA12-01_2024_Oct_Q06

6. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.
Using the laws of logarithms, solve

$$\log_4(12 - 2x) = 2 + 2\log_4(x + 1)$$

(5)

T09 Laws of Logarithms

09_LAWS Log laws

09_LAWS_CORE core family
move

WORKED SOLUTION

Question 6

WMA12/01 October 2024

WMA12-01_2024_Oct_Q06

1

State the domain

The logarithms require $12 - 2x > 0$ and $x + 1 > 0$, so $-1 < x < 6$.

2

Combine the right hand side

$2 = \log_4(16)$, so $2 + 2\log_4(x + 1) = \log_4(16(x + 1)^2)$.

3

Equate the arguments

$\log_4(12 - 2x) = \log_4(16(x + 1)^2)$ gives $12 - 2x = 16(x + 1)^2$.

4

Form the quadratic

$12 - 2x = 16x^2 + 32x + 16$, so $8x^2 + 17x + 2 = 0$.

5

Factor and solve

$8x^2 + 17x + 2 = (8x + 1)(x + 2) = 0$, so $x = -\frac{1}{8}$ or $x = -2$.

6

Check the domain

The value $x = -2$ is outside $-1 < x < 6$, so the solution is $x = -\frac{1}{8}$.

FINAL ANSWER

$$x = -\frac{1}{8}.$$

WMA12-01_2024_Oct_Q06

TOPIC 09 | FAMILY 04

Log Equations

WHAT THIS FAMILY ASKS

Recognise the log equations demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 Quadratics from logs can create invalid negative arguments.

15 classified questions

114 total marks

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09_SOLVE_CORE

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8 marks

WMA12/01 JANUARY 2021

Question 3

Past-paper question

WMA12-01_2021_Jan_Q03

3. (i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

(ii) Using the laws of logarithms, solve

$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

T09 Laws of Logarithms

09_SOLVE Log equations

09_SOLVE_CORE core family
move

WORKED SOLUTION

Question 3

WMA12/01 January 2021

WMA12-01_2021_Jan_Q03

1

Rewrite the exponential equationFrom $7^{x+2} = 3$, divide by 7^2 to get $7^x = \frac{3}{49}$.

2

Write x as a logarithmTherefore $x = \log_7 \left(\frac{3}{49} \right)$.

3

Convert one to a logarithmSince $1 = \log_2 2$, the equation becomes $\log_2 2 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$.

4

Combine the logarithms $\log_2 \{2y(y + 4)\} = \log_2 (5 - y)$.

5

Solve the quadraticThus $2y(y + 4) = 5 - y$, so $2y^2 + 9y - 5 = 0 = (2y - 1)(y + 5)$.

6

Check the domainThe original logarithms require $y > 0$ and $y < 5$, so $y = -5$ is invalid and $y = \frac{1}{2}$.**FINAL ANSWER**(i) $x = \log_7 \left(\frac{3}{49} \right)$. (ii) $y = \frac{1}{2}$.

WMA12-01_2021_Jan_Q03

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8 marks

WMA12/01 MAY/JUNE 2021

Question 2

Past-paper question

WMA12-01_2021_MayJune_Q02

2.

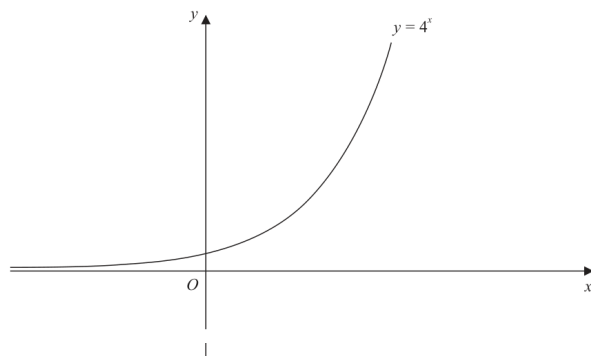


Figure 1

Figure 1 shows a sketch of the curve with equation $y = 4^x$.

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(a) On Diagram 1, sketch the curve with equation

(i) $y = 2^x$

(ii) $y = 4^x - 6$

Label clearly the coordinates of any points of intersection with the coordinate axes.

(4)

The curve with equation $y = 2^x$ meets the curve with equation $y = 4^x - 6$ at the point P .

(b) Using algebra, find the exact coordinates of P .

(4)

Question 2 continued

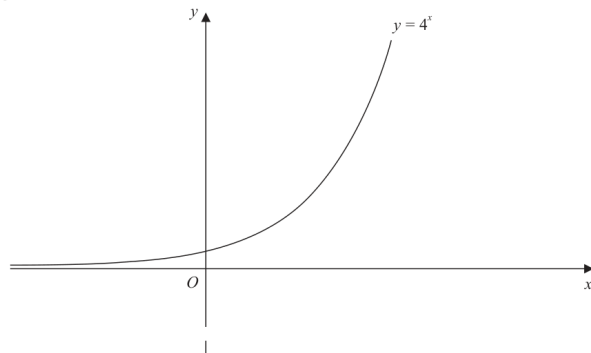


Diagram 1

T09 Laws of Logarithms

09_SOLVE Log equations

09_SOLVE_CORE core family
move

WORKED SOLUTION

Question 2

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q02

1

Describe y equals two to the x

$y = 2^x$ is an increasing exponential curve. It crosses the y -axis at $(0, 1)$ and has no x -axis intercept.

2

Describe y equals four to the x minus six

$y = 4^x - 6$ is the graph of $y = 4^x$ translated 6 units down. It crosses the y -axis at $(0, -5)$.

3

Find the x -axis intercept

For $y = 4^x - 6$, set $4^x - 6 = 0$. Hence $4^x = 6$ and $x = \log_4 6$.

4

Set up the intersection equation

At P , $2^x = 4^x - 6$.

5

Use a substitution

Let $u = 2^x$. Then $4^x = (2^x)^2 = u^2$, so $u = u^2 - 6$.

6

Solve for u

$u^2 - u - 6 = 0$, so $(u - 3)(u + 2) = 0$. Since $u = 2^x > 0$, $u = 3$.

7

State P

$2^x = 3$, so $x = \log_2 3$, and $y = 3$. Therefore $P = (\log_2 3, 3)$.

FINAL ANSWER

WMA12-01_2021_MayJune_Q02

(a)(i) Sketch $y = 2^x$ through $(0, 1)$ with no x -axis intercept. (a)(ii) Sketch $y = 4^x - 6$ through $(0, -5)$ and $(\log_4 6, 0)$. (b) $P = (\log_2 3, 3)$.

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5 marks

WMA12/01 JANUARY 2022

Question 4

Past-paper question

WMA12-01_2022_Jan_Q04

4. Using the laws of logarithms, solve

$$\log_3(32 - 12x) = 2\log_3(1 - x) + 3$$

(5)

DR ESLAM AHMED

T09 Laws of Logarithms

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WORKED SOLUTION

Question 4

WMA12/01 January 2022

WMA12-01_2022_Jan_Q04

1

Rewrite the right side $2 \log_3(1 - x) = \log_3(1 - x)^2$ and $3 = \log_3 27$.

2

Combine the logarithmsThe equation becomes $\log_3(32 - 12x) = \log_3\{27(1 - x)^2\}$.

3

Equate the arguments $32 - 12x = 27(1 - x)^2$.

4

Form the quadratic $32 - 12x = 27 - 54x + 27x^2$, so $27x^2 - 42x - 5 = 0$.

5

Solve $x = \frac{42 \pm 48}{54}$, giving $x = \frac{5}{3}$ or $x = -\frac{1}{9}$.

6

Check the domainThe original logarithms require $1 - x > 0$, so $x < 1$. Therefore $x = \frac{5}{3}$ is invalid and $x = -\frac{1}{9}$.**FINAL ANSWER**

$$x = -\frac{1}{9}.$$

WMA12-01_2022_Jan_Q04

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6 marks

WMA12/01 MAY/JUNE 2022

Question 4

Past-paper question

WMA12-01_2022_MayJune_Q04

4. In this question you must show all stages of your working.

Give your answers in fully simplified surd form.

Given that a and b are positive constants, solve the simultaneous equations

$$\begin{aligned}a - b &= 8 \\ \log_4 a + \log_4 b &= 3\end{aligned}$$

(6)

T09 Laws of Logarithms

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WORKED SOLUTION

Question 4

WMA12/01 May/June 2022

WMA12-01_2022_MayJune_Q04

1

Combine the logarithms

$$\log_4 a + \log_4 b = \log_4(ab) = 3.$$

2

Find ab

$$\text{Therefore } ab = 4^3 = 64.$$

3

Use a minus b

$$\text{From } a - b = 8, b = a - 8.$$

4

Form a quadratic

$$a(a - 8) = 64, \text{ so } a^2 - 8a - 64 = 0.$$

5

Solve for a

$$a = \frac{8 \pm \sqrt{64 + 256}}{2} = 4 \pm 4\sqrt{5}.$$

6

Choose positive constants

$$\text{Since } b = a - 8 \text{ must also be positive, take } a = 4 + 4\sqrt{5}.$$

7

Find b

$$b = a - 8 = 4\sqrt{5} - 4.$$

FINAL ANSWER

$$a = 4 + 4\sqrt{5}, \quad b = 4\sqrt{5} - 4.$$

WMA12-01_2022_MayJune_Q04

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6 marks

WMA12/01 JANUARY 2023

Question 4

Past-paper question

WMA12-01_2023_Jan_Q04

4. (i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7) \quad (3)$$

- (ii) Given that

$$\sum_{r=1}^2 \log_a(y^r) = \sum_{r=1}^2 (\log_a y)^r \quad y > 1, a > 1, y \neq a$$

find y in terms of a , giving your answer in simplest form. (3)

T09 Laws of Logarithms

09_SOLVE Log equations

09_SOLVE_CORE core family
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WORKED SOLUTION

Question 4

WMA12/01 January 2023

WMA12-01_2023_Jan_Q04

1

Rewrite the constant

Since $2 = \log_3 9$, the equation becomes $\log_3(4x) + \log_3 9 = \log_3(5x + 7)$.

2

Combine the logarithms

$\log_3(4x) + \log_3 9 = \log_3(36x)$.

3

Solve for x

Therefore $36x = 5x + 7$, so $31x = 7$ and $x = \frac{7}{31}$.

4

Introduce L

Let $L = \log_a y$. Then $\log_a(y^r) = r \log_a y = rL$.

5

Evaluate the left sum

$\sum_{r=1}^2 \log_a(y^r) = L + 2L = 3L$.

6

Evaluate the right sum

$\sum_{r=1}^2 (\log_a y)^r = L + L^2$.

7

Solve for L

So $3L = L + L^2$, which gives $L^2 - 2L = 0$ and $L(L - 2) = 0$.

8

Use the restrictions

Since $a > 1$ and $y > 1$, $\log_a y > 0$, so $L \neq 0$. Hence $L = 2$ and $y = a^2$.

FINAL ANSWER

(i) $x = \frac{7}{31}$. (ii) $y = a^2$.

WMA12-01_2023_Jan_Q04

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6 marks

WMA12/01 OCTOBER 2023

Question 5

Past-paper question

WMA12-01_2023_Oct_Q05

5. In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

(i) Solve

$$3^a = 70$$

giving the answer to 3 decimal places.

(2)

(ii) Find the exact value of b such that

$$4 + 3 \log_3 b = \log_3 5b$$

(4)

T09 Laws of Logarithms

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WORKED SOLUTION

Question 5

WMA12/01 October 2023

WMA12-01_2023_Oct_Q05

1

Solve the exponential equation

$$3^a = 70 \text{ gives } a = \log_3 70 = \frac{\log 70}{\log 3}.$$

2

Round a $a = 3.867147 \dots$, so $a = 3.867$ to 3 decimal places.

3

Rewrite four as a logarithm

$$4 = \log_3(3^4) = \log_3 81.$$

4

Combine the left side

$$4 + 3 \log_3 b = \log_3 81 + \log_3(b^3) = \log_3(81b^3).$$

5

Equate argumentsThe equation becomes $\log_3(81b^3) = \log_3(5b)$, so $81b^3 = 5b$.

6

Use the log domainSince $\log_3 b$ is defined, $b > 0$. Divide by b to get $81b^2 = 5$.

7

Find b

$$b^2 = \frac{5}{81}, \text{ and } b > 0, \text{ so } b = \frac{\sqrt{5}}{9}.$$

FINAL ANSWER

WMA12-01_2023_Oct_Q05

$$(i) a = 3.867. \quad (ii) b = \frac{\sqrt{5}}{9}.$$

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8 marks

WMA12/01 JANUARY 2024

Question 6

Past-paper question

WMA12-01_2024_Jan_Q06

6. (a) Given that

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2}$$

show that

$$x^3 + 6x^2 + x - 4 = 0 \quad (4)$$

(b) Given also that -1 is a root of the equation

$$x^3 + 6x^2 + x - 4 = 0$$

(i) use algebra to find the other two roots of the equation. (3)

(ii) Hence solve

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2} \quad (1)$$

T09 Laws of Logarithms

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WORKED SOLUTION

Question 6

WMA12/01 January 2024

WMA12-01_2024_Jan_Q06

1

Combine the logarithms $2 \log_4(x+3) = \log_4((x+3)^2)$, so the left side is $\log_4(x(x+3)^2)$.

2

Rewrite one half $\frac{1}{2} = \log_4 2$, so the right side is $\log_4(4x+2) + \log_4 2 = \log_4(8x+4)$.

3

Equate argumentsTherefore $x(x+3)^2 = 8x+4$.

4

Expand $x(x^2+6x+9) = 8x+4$, so $x^3+6x^2+9x = 8x+4$.

5

Show the cubicRearranging gives $x^3+6x^2+x-4 = 0$.

6

Use the given rootSince -1 is a root, $(x+1)$ is a factor of x^3+6x^2+x-4 .

7

Factor the cubic $x^3+6x^2+x-4 = (x+1)(x^2+5x-4)$.

8

Solve the quadratic $x^2+5x-4 = 0$ gives $x = \frac{-5 \pm \sqrt{25+16}}{2} = \frac{-5 \pm \sqrt{41}}{2}$.

9

Apply the logarithm domainThe original log equation requires $x > 0$. The only positive root is $x = \frac{-5+\sqrt{41}}{2}$.**FINAL ANSWER**

WMA12-01_2024_Jan_Q06

(a) **Proven.** (b)(i) $x = \frac{-5+\sqrt{41}}{2}$ and $x = \frac{-5-\sqrt{41}}{2}$. (b)(ii) $x = \frac{-5+\sqrt{41}}{2}$.

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9 marks

WMA12/01 MAY/JUNE (R) 2024

Question 9

Past-paper question

WMA12-01_2024_MayJuneR_Q09

9. Given that

$$3 \log_2(t+4) - 2 \log_2(t-2) = 7$$

(a) verify that $t = 4$ is a solution of the above equation,

(2)

(b) show that

$$t^3 - 116t^2 + 560t - 448 = 0$$

(3)

(c) Hence, using algebra and showing your working, solve

$$3 \log_2(t+4) - 2 \log_2(t-2) = 7$$

giving each answer in simplest form.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

T09 Laws of Logarithms

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WORKED SOLUTION

Question 9

WMA12/01 May/June (R) 2024

WMA12-01_2024_MayJuneR_Q09

1

Verify t equals fourSubstitute $t = 4$: $3 \log_2(8) - 2 \log_2(2) = 3(3) - 2(1) = 7$.

2

Combine the logarithms

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = \log_2 \left(\frac{(t+4)^3}{(t-2)^2} \right).$$

3

Use seven as a logarithm

$$7 = \log_2(128), \text{ so } \frac{(t+4)^3}{(t-2)^2} = 128.$$

4

Clear the denominator

$$(t + 4)^3 = 128(t - 2)^2.$$

5

Expand

$$t^3 + 12t^2 + 48t + 64 = 128t^2 - 512t + 512.$$

6

Show the cubic

$$\text{Rearranging gives } t^3 - 116t^2 + 560t - 448 = 0.$$

7

Use the known rootSince $t = 4$ is a root, $(t - 4)$ is a factor.

8

Factor the cubic

$$t^3 - 116t^2 + 560t - 448 = (t - 4)(t^2 - 112t + 112).$$

9

Solve the quadratic

$$t^2 - 112t + 112 = 0 \text{ gives } t = \frac{112 \pm \sqrt{112^2 - 4(112)}}{2} = 56 \pm 12\sqrt{21}.$$

10

Use the log domain

The original equation requires $t > 2$. The value $56 - 12\sqrt{21}$ is less than 2, so it is rejected.
Hence $t = 4$ or $t = 56 + 12\sqrt{21}$.

FINAL ANSWER

WMA12-01_2024_MayJuneR_Q09

(a) **Verified.** (b) **Proven.** (c) $t = 4$ **or** $t = 56 + 12\sqrt{21}$.

DR ESLAM AHMED

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8 marks

WMA12/01 JANUARY 2025

Question 7

Past-paper question

WMA12-01_2025_Jan_Q07

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) The table below shows values of x and y , where $y = \log_{10}(x + 5)$, for x values between -1 and 4

x	-1	0	1	2	3	4
$y = \log_{10}(x + 5)$	$\log_{10} 4$	$\log_{10} 5$	$\log_{10} 6$	$\log_{10} 7$	$\log_{10} 8$	$\log_{10} 9$

Using the trapezium rule with all the y values in the given table, show that

$$\int_{-1}^4 \log_{10}(x + 5) \, dx \approx \log_{10} k$$

where k is an integer to be found.

(3)

- (ii) Find the value of a such that

$$2\log_5(5 - a) - \log_5(a + 25) = 1$$

(5)

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WORKED SOLUTION

Question 7

WMA12/01 January 2025

WMA12-01_2025_Jan_Q07

1

Use the trapezium rule

The width is 1, so $\int_{-1}^4 \log_{10}(x+5) dx \approx \frac{1}{2} \{ \log_{10} 4 + \log_{10} 9 + 2(\log_{10} 5 + \log_{10} 6 + \log_{10} 7 + \log_{10} 8) \}$.

2

Combine the logarithms

This is $\log_{10} (\sqrt{4 \times 9} \times 5 \times 6 \times 7 \times 8)$.

3

Find k

$\sqrt{4 \times 9} \times 5 \times 6 \times 7 \times 8 = 6 \times 1680 = 10080$, so $k = 10080$.

4

State the domain

The logarithms require $5 - a > 0$ and $a + 25 > 0$, so $-25 < a < 5$.

5

Combine the logarithms

$2 \log_5 (5 - a) - \log_5 (a + 25) = \log_5 \left(\frac{(5-a)^2}{a+25} \right)$.

6

Use one as a logarithm

$1 = \log_5 5$, so $\frac{(5-a)^2}{a+25} = 5$.

7

Form the quadratic

$(5 - a)^2 = 5(a + 25)$, so $a^2 - 10a + 25 = 5a + 125$. Hence $a^2 - 15a - 100 = 0$.

8

Solve and check

$(a - 20)(a + 5) = 0$, so $a = 20$ or $a = -5$. Only $a = -5$ is in the domain, so $a = -5$.

FINAL ANSWER

(i) $k = 10080$. (ii) $a = -5$.

WMA12-01_2025_Jan_Q07

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10 marks

WMA12/01 MAY/JUNE 2025

Question 6

Past-paper question

WMA12-01_2025_MayJune_Q06

6:

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right) \quad (5)$$

(ii) The curves C_1 and C_2 with equations

$$y = 3^{2x+1} \quad \text{and} \quad y = 6 \times 3^x$$

meet at the point P .

Find the exact coordinates of P , writing your answer in the form $(\log_3 a, b)$ where a and b are integers.

(5)

T09 Laws of Logarithms

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WORKED SOLUTION

Question 6

WMA12/01 May/June 2025

WMA12-01_2025_MayJune_Q06

1

State the domain

The logarithms require $4 - x > 0$ and $\frac{x+11}{2} > 0$, so $-11 < x < 4$.

2

Use log laws

$2 \log_2(4 - x) = \log_2((4 - x)^2)$ and $3 + \log_2\left(\frac{x+11}{2}\right) = \log_2\left(8 \cdot \frac{x+11}{2}\right)$.

3

Equate the arguments

Thus $(4 - x)^2 = 4(x + 11)$.

4

Solve the quadratic

$x^2 - 8x + 16 = 4x + 44$, so $x^2 - 12x - 28 = 0$. Hence $(x - 14)(x + 2) = 0$.

5

Check the domain

The root 14 is outside the domain, so $x = -2$.

6

Equate the curves

At P , $3^{2x+1} = 6 \times 3^x$.

7

Let u equal 3 to the x

Let $u = 3^x$. Then $3^{2x+1} = 3u^2$, so $3u^2 = 6u$.

8

Find u

Since $u > 0$, divide by $3u$ to get $u = 2$. Hence $3^x = 2$ and $x = \log_3 2$.

9

Find y

$y = 6 \times 3^x = 6 \times 2 = 12$, so $P = (\log_3 2, 12)$.

FINAL ANSWER

WMA12-01_2025_MayJune_Q06

(i) $x = -2$. (ii) $P = (\log_3 2, 12)$.

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WMA12/01 OCTOBER 2025

Question 9

Past-paper question

WMA12-01_2025_Oct_Q09

9. **In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_3(4x + 5) - \log_3(x + 3) = 2 \quad (5)$$

(ii) Given that $a > 0$, $b > 0$ and

$$\log_{10} a + \log_{10} b = \log_{10}(a + b)$$

(a) prove that $a = \frac{b}{b-1}$ (3)

(b) Hence write down the full restriction on the value of b , giving a reason for your answer. (2)

T09 Laws of Logarithms

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WORKED SOLUTION

Question 9

WMA12/01 October 2025

WMA12-01_2025_Oct_Q09

1

State the domainThe logarithms require $4x + 5 > 0$ and $x + 3 > 0$, so $x > -\frac{5}{4}$.

2

Combine the logarithms

$$2\log_3(4x + 5) - \log_3(x + 3) = \log_3\left(\frac{(4x+5)^2}{x+3}\right).$$

3

Use two as a logarithm

$$2 = \log_3 9, \text{ so } \frac{(4x+5)^2}{x+3} = 9.$$

4

Form the quadratic

$$(4x + 5)^2 = 9(x + 3), \text{ so } 16x^2 + 40x + 25 = 9x + 27. \text{ Hence } 16x^2 + 31x - 2 = 0.$$

5

Solve the quadratic

$$16x^2 + 31x - 2 = (16x - 1)(x + 2) = 0, \text{ so } x = \frac{1}{16} \text{ or } x = -2.$$

6

Check the domain $x = -2$ is outside the domain, so $x = \frac{1}{16}$.

7

Use the logarithm equation

$$\log_{10} a + \log_{10} b = \log_{10}(a + b) \text{ gives } \log_{10}(ab) = \log_{10}(a + b).$$

8

Equate the argumentsSince the logarithm function is one-to-one, $ab = a + b$.

9

Rearrange for a

$$ab - a = b, \text{ so } a(b - 1) = b \text{ and therefore } a = \frac{b}{b-1}.$$

10

Restrict bGiven $a > 0$ and $b > 0$, the fraction $\frac{b}{b-1}$ must be positive. Since the numerator is positive, the denominator must also be positive, so $b > 1$.

FINAL ANSWER

WMA12-01_2025_Oct_Q09

(i) $x = \frac{1}{16}$. (ii)(a) **Proven.** (ii)(b) $b > 1$.

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10 marks

WMA12/01 MAY/JUNE 2021

Question 7

Past-paper question

WMA12-01_2021_MayJune_Q07

7. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0$$

(4)

- (b) Show that -4 is a root of this cubic equation.

(2)

- (c) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

(4)

T09 Laws of Logarithms

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09_SOLVE_HV02 Quadratics from logs can create invalid negative arguments.

WORKED SOLUTION

Question 7

WMA12/01 May/June 2021

WMA12-01_2021_MayJune_Q07

1

Use the power law

$$3 \log_3(2x - 1) = \log_3(2x - 1)^3.$$

2

Rewrite two as a logarithm

Since $2 = \log_3 9$, the right side is $\log_3 9 + \log_3(14x - 25) = \log_3\{9(14x - 25)\}$.

3

Equate the arguments

So $(2x - 1)^3 = 9(14x - 25)$.

4

Expand and simplify

$8x^3 - 12x^2 + 6x - 1 = 126x - 225$, which simplifies to $2x^3 - 3x^2 - 30x + 56 = 0$.

5

Check the given root

Substituting $x = -4$ gives $2(-64) - 3(16) - 30(-4) + 56 = 0$, so -4 is a root.

6

Factorise the cubic

Dividing by $(x + 4)$ gives $2x^2 - 11x + 14 = (2x - 7)(x - 2)$.

7

Apply the logarithm domain

The possible roots are -4 , 2 , and $\frac{7}{2}$. The original logarithms require $2x - 1 > 0$ and $14x - 25 > 0$, so $x > \frac{25}{14}$.

8

State the valid solutions

Therefore the solutions are $x = 2$ and $x = \frac{7}{2}$.

FINAL ANSWER

WMA12-01_2021_MayJune_Q07

(a) Proven. (b) -4 is a root of the cubic. (c) $x = 2, \frac{7}{2}$.

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5 marks

WMA12/01 MAY/JUNE 2023

Question 5

Past-paper question

WMA12-01_2023_MayJune_Q05

5. Use the laws of logarithms to solve

$$\log_2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6)$$

(5)

T09 Laws of Logarithms

09_SOLVE Log equations

09_SOLVE_HV02 Quadratics
from logs can create invalid
negative arguments.

WORKED SOLUTION

Question 5

WMA12/01 May/June 2023

WMA12-01_2023_MayJune_Q05

1

State the domain

The logarithms require $16x > 0$, $x + 1 > 0$ and $x + 6 > 0$, so $x > 0$.

2

Combine the left side

$$\log_2(16x) + \log_2(x + 1) = \log_2(16x(x + 1)).$$

3

Rewrite the right side

$$3 = \log_2 8, \text{ so } 3 + \log_2(x + 6) = \log_2(8(x + 6)).$$

4

Equate the arguments

$$\text{Therefore } 16x(x + 1) = 8(x + 6).$$

5

Solve the quadratic

$$16x^2 + 16x = 8x + 48, \text{ so } 16x^2 + 8x - 48 = 0. \text{ Dividing by 8 gives } 2x^2 + x - 6 = 0.$$

6

Factor and use the domain

$$(2x - 3)(x + 2) = 0, \text{ so } x = \frac{3}{2} \text{ or } x = -2. \text{ The domain } x > 0 \text{ rejects } -2, \text{ leaving } x = \frac{3}{2}.$$

FINAL ANSWER

$$x = \frac{3}{2}.$$

WMA12-01_2023_MayJune_Q05

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6 marks

WMA12/01 MAY/JUNE 2024

Question 3

Past-paper question

WMA12-01_2024_MayJune_Q03

3. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Using the laws of logarithms, solve

$$2 \log_2 (2 - x) = 4 + \log_2 (x + 10) \quad (5)$$

- (ii) Find the value of

$$\log_{\sqrt{a}} a^6$$

where a is a positive constant greater than 1

(1)

T09 Laws of Logarithms

09_SOLVE Log equations

09_SOLVE_HV02 Quadratics
from logs can create invalid
negative arguments.

WORKED SOLUTION

Question 3

WMA12/01 May/June 2024

WMA12-01_2024_MayJune_Q03

1

State the domain

The logarithms require $2 - x > 0$ and $x + 10 > 0$, so $-10 < x < 2$.

2

Combine powers

$$2 \log_2(2 - x) = \log_2((2 - x)^2).$$

3

Rewrite four

$$4 = \log_2 16, \text{ so } 4 + \log_2(x + 10) = \log_2(16(x + 10)).$$

4

Equate arguments

$$(2 - x)^2 = 16(x + 10).$$

5

Solve the quadratic

$$x^2 - 4x + 4 = 16x + 160, \text{ so } x^2 - 20x - 156 = 0.$$

6

Use the domain

$x = 26$ or $x = -6$. The value 26 is outside the domain, so $x = -6$.

7

Evaluate the logarithm

$$\log_{\sqrt{a}}(a^6) = \log_{a^{1/2}}(a^6) = \frac{6}{1/2} = 12.$$

FINAL ANSWER

(i) $x = -6$. (ii) 12.

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9 marks

WMA12/01 JANUARY 2026

Question 7

Past-paper question

WMA12-01_2026_Jan_Q07

7. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

- (i) Find the exact solution of the equation

$$8^{2x-5} = 20$$

giving the answer in the form $a + b \log_2 5$ where a and b are rational constants.

(4)

- (ii) Using the laws of logarithms, solve the equation

$$\log_3 (13 + 2y) + 3 = 2 \log_3 (4 - y)$$

(5)

T09 Laws of Logarithms

09_SOLVE Log equations

09_SOLVE_HV02 Quadratics from logs can create invalid negative arguments.

WORKED SOLUTION

Question 7

WMA12/01 January 2026

WMA12-01_2026_Jan_Q07

1

Take logarithms base 2

From $8^{2x-5} = 20$, take $\log_2$ of both sides to get $(2x - 5) \log_2 8 = \log_2 20$.

2

Simplify the logarithms

Since $\log_2 8 = 3$ and $\log_2 20 = \log_2(4 \cdot 5) = 2 + \log_2 5$, we get $3(2x - 5) = 2 + \log_2 5$.

3

Solve for x

$6x - 15 = 2 + \log_2 5$, so $6x = 17 + \log_2 5$ and $x = \frac{17}{6} + \frac{1}{6} \log_2 5$.

4

Use log laws

$3 = \log_3 27$ and $2 \log_3(4 - y) = \log_3(4 - y)^2$.

5

Remove the logarithms

$\log_3(13 + 2y) + \log_3 27 = \log_3(4 - y)^2$, so $27(13 + 2y) = (4 - y)^2$.

6

Solve the quadratic

$351 + 54y = y^2 - 8y + 16$, hence $y^2 - 62y - 335 = 0 = (y - 67)(y + 5)$.

7

Check the logarithm domains

We need $13 + 2y > 0$ and $4 - y > 0$. Therefore $y = 67$ is not valid, and the solution is $y = -5$.

FINAL ANSWER

WMA12-01_2026_Jan_Q07

(i) $x = \frac{17}{6} + \frac{1}{6} \log_2 5$. (ii) $y = -5$.